

AI-Assisted Threat Detection Dashboard

Introduction

1.1 Overview

Cybersecurity has become one of the most critical aspects of modern organizations due to the increasing number of cyberattacks, ransomware incidents, phishing campaigns, insider threats, and advanced persistent threats (APTs). Organizations generate millions of security events every day from firewalls, servers, endpoints, cloud platforms, and applications. Manual monitoring of these events is time-consuming and often results in delayed threat detection.

The **AI-Assisted Threat Detection Dashboard** is designed to automate cybersecurity monitoring using Artificial Intelligence (AI), Machine Learning (ML), and Threat Intelligence. The platform continuously analyzes security logs, detects abnormal behavior, calculates risk scores, and provides security analysts with real-time dashboards for faster incident response.

Key Points

- AI-based cybersecurity monitoring
- Real-time threat detection
- Machine learning for anomaly detection
- Automated risk scoring
- Interactive dashboard
- Threat intelligence integration
- Security reporting and visualization

1.2 Need for the Project

Organizations face thousands of cyber threats every day. Existing systems generate many false alerts, making it difficult for analysts to identify real attacks.

Need

- Reduce manual monitoring efforts.
- Detect threats automatically.
- Improve incident response time.
- Reduce false-positive alerts.
- Provide centralized threat visibility.
- Enhance cybersecurity posture.

1.3 Problem Statement

Traditional security monitoring systems depend mainly on rule-based detection mechanisms, which cannot identify unknown or zero-day attacks effectively. Security analysts spend considerable time investigating false alarms, leading to delayed responses and increased operational costs.

The proposed system addresses these limitations by using machine learning models to detect abnormal activities, prioritize incidents, and assist analysts in making informed security decisions.

OBJECTIVES

Primary Objectives

- Develop an AI-powered threat detection platform.
- Collect security logs from multiple sources.
- Detect anomalous activities using ML algorithms.
- Integrate threat intelligence feeds.
- Calculate cybersecurity risk scores.
- Display real-time dashboards.
- Generate automated reports.

Secondary Objectives

- Improve detection accuracy.
- Reduce false alarms.
- Support decision-making.
- Enable predictive threat analysis.
- Improve operational efficiency.

CHAPTER 2: SCOPE

In Scope

- Security log analysis
- Threat intelligence integration
- Risk scoring
- Dashboard visualization
- Security analytics
- Incident prioritization
- Report generation

Out of Scope

- Malware removal
- Automatic firewall configuration
- Penetration testing
- Digital forensics
- Network device management

CHAPTER 3: EXISTING SYSTEM

Existing Methods

- Manual log monitoring
- SIEM tools
- Rule-based detection
- Signature-based IDS

Limitations

- High false positives
- No AI intelligence
- Slow response
- Large alert volume
- Limited visualization
- Difficult prioritization
- High operational cost

CHAPTER 4: PROPOSED SYSTEM

The proposed AI-Assisted Threat Detection Dashboard combines Artificial Intelligence, Machine Learning, and Cyber Threat Intelligence into a centralized platform capable of identifying sophisticated cyber threats.

Features

- Real-time monitoring
- Threat intelligence integration
- AI anomaly detection
- Risk prioritization
- Interactive dashboard
- Executive reporting
- Predictive analytics

Advantages

- Faster threat detection
- Reduced workload
- Improved accuracy
- Better decision-making
- Scalable architecture
- Real-time visualization

CHAPTER 5: SYSTEM REQUIREMENTS

Hardware Requirements

- Intel Core i5 Processor or above
- 8 GB RAM (Minimum)
- 256 GB SSD
- Internet Connection

Software Requirements

- Windows/Linux
- Python 3.11+
- MongoDB
- VS Code
- Jupyter Notebook
- Dash
- Plotly
- Pandas
- NumPy
- Scikit-learn

CHAPTER 6: SYSTEM ARCHITECTURE

Components

Data Sources

- Firewall Logs
- IDS Logs
- Authentication Logs
- Server Logs
- Threat Intelligence Feeds
- CVE Database

Data Collection Layer

- Log Collection
- Data Validation
- Data Cleaning

AI Engine

- Feature Extraction
- Anomaly Detection
- Classification
- Prediction

Risk Scoring Engine

- Threat Confidence
- Asset Criticality
- Vulnerability Score
- Risk Prioritization

Dashboard

- Threat Trends
- Charts
- Reports
- Alerts

CHAPTER 7: MODULES

Module 1

Security Data Aggregation

Functions

- Log Collection
- Data Cleaning
- Data Normalization
- Threat Intelligence Mapping
- Data Validation

Deliverables

- Processed datasets
- Standardized logs
- Threat mapping

Module 2

AI-Based Threat Detection

Functions

- Feature Engineering
- Anomaly Detection
- Classification
- Prediction
- Threat Confidence Score

Algorithms

- Isolation Forest
- Random Forest

- Logistic Regression
- Decision Tree

Module 3

Risk Prioritization

Functions

- Risk Calculation
- Severity Classification
- Threat Correlation
- Response Recommendation

Risk Levels

- Low
- Medium
- High
- Critical

Module 4

Dashboard

Dashboard Pages

- Home
- Live Monitoring
- Threat Analytics
- Risk Analysis
- Reports
- Admin Panel

Dashboard Features

- Line Charts

- Bar Charts
- Pie Charts
- Heat Maps
- Filters
- Drill Down Analysis

CHAPTER 8: DATABASE DESIGN

Collections

Users

- User ID
- Username
- Email
- Password
- Role

Security Logs

- Log ID
- Timestamp
- Source IP
- Destination IP
- Event Type
- Severity
- Status

Threat Intelligence

- Threat ID
- CVE ID
- MITRE Technique
- Threat Category
- Confidence Score

Risk Assessment

- Incident ID
- Risk Score
- Priority
- Recommendation

CHAPTER 9: MACHINE LEARNING

Data Preprocessing

- Missing Value Handling
- Data Cleaning
- Feature Selection
- Normalization
- Encoding

Model Training

- Training Dataset
- Testing Dataset
- Cross Validation
- Hyperparameter Tuning

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- ROC Curve
- Confusion Matrix

CHAPTER 10: TESTING

Types

- Unit Testing
- Integration Testing
- Functional Testing
- Security Testing
- Performance Testing
- User Acceptance Testing

Test Cases

- Login
- Log Upload
- Threat Detection
- Dashboard
- Report Generation
- Alert Notification

CHAPTER 11: RESULTS

The system successfully:

- Aggregated logs from multiple sources.
- Identified anomalous activities.
- Generated threat confidence scores.
- Calculated cybersecurity risk scores.
- Prioritized incidents.
- Displayed threat trends.
- Generated executive reports.

Performance

- Threat detection accuracy improved.
- False positives reduced.
- Faster incident response.
- Improved analyst productivity.

CHAPTER 12: FUTURE ENHANCEMENTS

- Deep Learning-based threat prediction
- Cloud deployment (AWS/Azure)
- SIEM integration
- Blockchain-based log security
- Automated incident response
- Mobile application
- Natural Language Processing for threat reports
- Generative AI assistant for SOC analysts

CHAPTER 14: CONCLUSION

The AI-Assisted Threat Detection Dashboard is a scalable cybersecurity solution that combines Artificial Intelligence, Machine Learning, and Threat Intelligence to improve cyber defense. It enables organizations to detect threats proactively, reduce false positives, prioritize incidents, and provide comprehensive security insights through an interactive dashboard. The project demonstrates how AI can significantly enhance cybersecurity operations and support informed decision-making in Security Operations Centers (SOCs).